PREDICTING EARNINGS MANIPULATIONS[1]

INTRODUCTION

Earnings manipulations imply that the company undertakes deliberate steps to bring reported earnings to a desired level. For example, companies can manipulate revenue, inventory and cost of goods sold. Accounting literature has proposed many models determining whether companies manipulate their earnings. For instance, Beneish[2] uses eight financial indices (Exhibit 1) to predict earnings manipulations by treating it as a classification problem.

In this assignment, a bank that provides business loans to small and medium enterprises suspects that its customers may be involved in earnings manipulations to increase their chances of securing a loan. The bank collected information on earnings manipulators and the Beneish model's eight variables.

DATA

The training data consists of 2490 data instances containing the eight variables of the Beneish model and corresponding manipulation labels. The data has 78/2490 = 3.13% positive samples (Manipulate==1). The data is stored in the 'training_data.csv' file.

The test data consists of 1227 data instances containing the eight variables of the Beneish model without the actual manipulation label. The percentage of positive samples (Manipulate==1) in this sample is approximately the same as that in the training data. The data is stored in the 'testing_data.csv' file.

OBJECTIVES

Part 1

In this part, we investigate whether we can predict earnings manipulations using machine learning. Keep in mind to deliver a structured notebook that follows the typical workflow of a data science project.

- First, examine the data and execute the necessary preprocessing steps. Next, develop a classification model to detect companies that manipulate their earnings. Notice that the data is imbalanced. In this case, traditional classification algorithms tend to be biased towards the majority class, which often results in unsatisfactory performance. The authors of the papers[3] [4] discuss additional methods suitable for imbalanced classification. Compare at least one baseline method with at least one method tailored for imbalanced classification.

[1] Unnikrishnan, D. K., Syed, T. A. I., Ganeshan , S., 2016, Predicting Earnings Manipulation by Indian Firms Using Machine Learning Algorithms, Harvard Business Review.

[2] Beneish, M. D., 1999, The detection of earnings manipulations, Financial Analyst Journal, 55(5), 24-36. doi: https://doi.org/10.2469/faj.v55.n5.2296

[3] Zhu, B., Baesens, B., vanden Broucke, S.K., 2017. An empirical comparison of techniques for the class imbalance problem in churn prediction. Information sciences 408, 84–99. doi: https://doi.org/10.1016/j.ins.2017.04.015

[4] Diez-Pastor, J.F., Rodriguez, J.J., Garcia-Osorio, C.I., Kuncheva, L.I., 2015. Diversity techniques improve the performance of the best imbalance learning ensembles. Information Sciences 325, 98–117. doi: https://doi.org/10.1016/j.ins.2015.07.025

For instance, compare logistic regression trained on the original data (baseline) with a logistic regression exploiting SMOTE samples (method for imbalanced classification). Discuss the choices that you made and explain why your model is suitable for this task.

- Compare your best model with the model that was proposed in the paper of Beneish[5]. Discuss the differences in methodology, performance, feature importance, …
- Submit your best model's prediction scores (decimal numbers between 0 and 1) on the test data in a .csv file. Your model will be assessed by computing the average precision (AP)[6] from your prediction scores. Denote this file as 'part_1_prediction_scores_your_last_names.csv'. Do not include a header and row indices. An example (for the artificial persons Olivia Smith and Dean Thomas) can be found in figure 1.

	A	B	C
1	0.013783053131324894		
2	0.016992820768917442		
3	0.018820766561633288		
4	0.025099808554182004		
5	0.018517358172823004		
6	0.02822945342607476		
7	0.03757686779187811		
8	0.020273772526586416		
9	0.02082169873003506		
10	0.01436584730807216		
11	0.03025555566847649		
12	0.036188416969670414		

(figure 1: structure of the file 'part_1_prediction_scores_smith_thomas.csv' file)

Part 2

In this part, we investigate how your previous findings translate into the risk of lending money to companies and the potential losses. Note that we only consider the losses here. We don't include the profit from the interest on loans. We assume the requested loans are typically about 1,000,000 dollars. Next, we assume that a company that manipulates its earnings to get this loan has a 10 per cent probability of going bankrupt and will be unable to repay it. Hence, when a company manipulates its earnings, but we fail to detect it, our expected loss is 100,000 dollars (1,000,000 dollars * 0.10). On the other hand, the cost of unnecessarily sending an accountant to check the bookkeeping of a non-fraudulent (the earnings are NOT manipulated) company is 5000 dollars.

- What is the expected loss of the model granting a loan to all instances in the training set? Next, what is the expected loss of your best model of the previous section on all instances in the training set? Compare the results.
- Would you develop a new model or alter your predictions after seeing this cost information? Explain (if applicable) the steps you took.

[5] Beneish, M. D., 1999, The detection of earnings manipulations, Financial Analyst Journal, 55(5), 24-36. doi: https://doi.org/10.2469/faj.v55.n5.2296

[6] https://scikit-learn.org/stable/modules/generated/sklearn.metrics.average_precision_score.html

- Submit your predictions on the test data in a .csv file. The predictions (binary!) should consist of 1's (grant no loan) and 0's (grant a loan). Your model will be assessed by computing the expected loss from your predictions. Denote this file as 'part_2_predictions_your_last_names.csv'. Do not include a header and row indices. An example (for the artificial persons Olivia Smith and Dean Thomas) can be found in figure 2.

	A	B	C
1	0		
2	0		
3	0		
4	0		
5	0		
6	0		
7	0		
8	0		
9	0		
10	0		
11	0		
12	0		

(figure 2: structure of the 'part_2_predictions_smith_thomas.csv' file)

Part 3

Prepare a presentation covering the material aspects of this assignment. Communicate your findings in a structured way.

For those wanting to go the extra mile…

Students who wish to go **beyond the core requirements** may choose to extend the assignment in one or more of the following ways.

Examples include (but are not limited to):

- Building a user-friendly dashboard (e.g. to visualize key patterns, risks, or outcome for a non-technical audience)
- Linking the case to more recent academic/practitioner research on fraud or machine-learning techniques relevant to fraud detection (e.g. anomaly detection)
- Obtaining additional data and using it to test or validate the model on new observations

Good luck!

EXHIBIT 1

Important note: Formulas might be incorrect in case file, here are the correct formulas:

$$\textbf{DSRI} = \frac{\text{Receivables}_t/\text{Sales}_t}{\text{Receivables}_{t-1}/\text{Sales}_{t-1}}$$

$$\textbf{GMI} = \frac{(\text{Sales}_{t-1} - \text{Cost of good sold}_{t-1})/\text{Sales}_{t-1}}{(\text{Sales}_t - \text{Cost of good sold}_t)/\text{Sales}_t}$$

$$\textbf{AQI} = \frac{1 - (\text{Current assets}_t + \text{PP\&E}_t)/\text{Total assets}_t}{1 - (\text{Current assets}_{t-1} + \text{PP\&E}_{t-1})/\text{Total assets}_{t-1}}$$

$$\textbf{SGI} = \frac{\text{Sales}_t}{\text{Sales}_{t-1}}$$

$$\textbf{DEPI} = \frac{\text{Depreciation}_{t-1}/(\text{Depreciation}_{t-1} + \text{PP\&E}_{t-1})}{\text{Depreciation}_t/(\text{Depreciation}_t + \text{PP\&E}_t)}$$

$$\textbf{SGAI} = \frac{\text{Sales, general, and administrative expense}_t/\text{Sales}_t}{\text{Sales, general, and administrative expense}_{t-1}/\text{Sales}_{t-1}}$$

$$\textbf{LVGI (or LEVI)} = \frac{(\text{LTD}_t + \text{Current liabilities}_t)/\text{Total assets}_t}{(\text{LTD}_{t-1} + \text{Current liabilities}_{t-1})/\text{Total assets}_{t-1}}$$

$$\textbf{TATA (or ACCR)} = \frac{\begin{array}{c}\Delta\text{Current assets}_t - \Delta\text{Cash}_t \\ - (\Delta\text{Current liabilities}_t - \Delta\text{Current maturities of LTD}_t - \Delta\text{Income tax payable}_t) \\ - \text{Depreciation and amortization}_t\end{array}}{\text{Total assets}_t}$$